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CONTROLLER, LASER DEVICE, AND ELECTRONIC DEVICE MANUFACTURING METHOD

Abstract

A controller includes a processor, a volatile memory, a nonvolatile memory, and a capacitor, and is configured to control a laser device. Here, the processor is configured to store a parameter of the laser device in the volatile memory when receiving power supply from outside the controller, and when the power supply from outside is stopped, store, in the nonvolatile memory, the parameter stored in the volatile memory and information indicating a duration time of power supply from the capacitor while receiving the power supply from the capacitor.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims the benefit of Japanese Patent Application No. 2024-028027, filed on Feb. 28, 2024, the entire contents of which are hereby incorporated by reference.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to a controller, a laser device, and an electronic device manufacturing method.

2. Related Art

[0003] Recently, in a semiconductor exposure apparatus, improvement in resolution has been desired for miniaturization and high integration of semiconductor integrated circuits. For this purpose, an exposure light source that outputs light having a shorter wavelength has been developed. For example, as a gas laser device for exposure, a KrF excimer laser device for outputting laser light having a wavelength of about 248 nm and an ArF excimer laser device for outputting laser light having a wavelength of about 193 nm are used.

[0004] The KrF excimer laser device and the ArF excimer laser device each have a large spectral line width of about 350 to 400 μm in natural oscillation light. Therefore, when a projection lens is formed of a material that transmits ultraviolet rays such as KrF laser light and ArF laser light, there is a case in which chromatic aberration occurs. As a result, the resolution may decrease. Then, a spectral line width of laser light output from the gas laser device needs to be narrowed to the extent that the chromatic aberration can be ignored. For this purpose, there is a case in which a line narrowing module (LNM) including a line narrowing element (etalon, grating, and the like) is provided in a laser resonator of the gas laser device to narrow a spectral line width. A gas laser device with a narrowed spectral line width is referred to as a line narrowing gas laser device.

LIST OF DOCUMENTS

Patent Documents

[0005] Patent Document 1: Japanese Patent Application Publication No. 2022-136817

SUMMARY

[0006] A controller according to an aspect of the present disclosure includes a processor, a volatile memory, nonvolatile memory, and a capacitor, and is configured to control a laser device. Here, the processor is configured to store a parameter of the laser device in the volatile memory when receiving power supply from outside the controller, and when the power supply from outside is stopped, store, in the nonvolatile memory, the parameter stored in the volatile memory and information indicating a duration time of power supply from the capacitor while receiving the power supply from the capacitor.

[0007] A laser device according to an aspect of the present disclosure includes an oscillator configured to output laser light, and a controller configured to control the oscillator. Here, the controller includes a processor, a volatile memory, a nonvolatile memory, and a capacitor. The processor is configured to store a parameter of the laser device in the volatile memory when receiving power supply from outside the controller, and when the power supply from outside is stopped, store, in the nonvolatile memory, the parameter stored in the volatile memory and information indicating a duration time of power supply from the capacitor while receiving the power supply from the capacitor.

[0008] An electronic device manufacturing method according to an aspect of the present disclosure includes generating laser light using a laser device, outputting the laser light to an exposure apparatus, and exposing a photosensitive substrate to the laser light in the exposure apparatus to manufacture an electronic device. Here, the laser device includes an oscillator configured to output

laser light, and a controller configured to control the oscillator. The controller includes a processor, a volatile memory, a nonvolatile memory, and a capacitor. The processor is configured to store a parameter of the laser device in the volatile memory when receiving power supply from outside the controller, and when the power supply from outside is stopped, store, in the nonvolatile memory, the parameter stored in the volatile memory and information indicating a duration time of power supply from the capacitor while receiving the power supply from the capacitor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] Embodiments of the present disclosure will be described below merely as examples with reference to the accompanying drawings.

[0010] FIG. 1 shows the configuration of an exposure system in a comparative example.

[0011] FIG. 2 shows the configuration of a laser device according to the comparative example.

[0012] FIG. 3 is a flowchart showing operation of a processor when an internal power source becomes off in a first embodiment.

[0013] FIG. 4 shows data of a part of a nonvolatile memory in the first embodiment.

[0014] FIG. 5 is a flowchart showing operation of the processor when the internal power source becomes on in the first embodiment.

[0015] FIG. 6 is a flowchart showing operation of the processor when the internal power source becomes off in a second embodiment.

[0016] FIG. 7 shows data of a part of the nonvolatile memory in the second embodiment.

DESCRIPTION OF EMBODIMENTS

<Contents>

[0017] 1. Comparative example [0018] 1.1 Configuration of exposure apparatus **200** [0019] 1.2 Operation of exposure apparatus **200** [0020] 1.3 Configuration of laser device **100** [0021] 1.4 Operation of laser device **100** [0022] 1.5 Problem of comparative example [0023] 2. Laser device **100** determining deterioration based on duration time of power supply from capacitor **24** [0024] 2.1 Operation when power source becomes off [0025] 2.2 Operation when power source becomes on [0026] 2.3 Effect [0027] 3. Laser device **100** sequentially storing information indicating duration time in different regions of nonvolatile memory **23** [0028] 3.1 Operation when power source becomes off [0029] 3.2 Operation when power source becomes on [0030] 3.3 Effect [0031] 4. Others

[0032] Hereinafter, embodiments of the present disclosure will be described in detail with reference to the drawings. The embodiments described below show some examples of the present disclosure and do not limit the contents of the present disclosure. Also, all configurations and operation described in the embodiments are not necessarily essential as configurations and operation of the present disclosure. Here, the same components are denoted by the same reference numeral, and duplicate description thereof is omitted.

1. Comparative Example

[0033] FIG. 1 shows the configuration of an exposure system in a comparative example. The comparative example of the present disclosure is an example recognized by the applicant as known only by the applicant, and is not a publicly known example admitted by the applicant.

[0034] The exposure system includes a laser device **100** and an exposure apparatus **200**. The laser device **100** is configured to output laser light LB toward the exposure apparatus **200**.

1.1 Configuration of Exposure Apparatus **200**

[0035] The exposure apparatus **200** includes an illumination optical system **201** and a projection optical system **202**. The illumination optical system **201** illuminates a reticle pattern of a reticle (not shown) arranged on a reticle stage RT with the laser light LB incident from the laser device

100. The projection optical system **202** causes the laser light LB transmitted through the reticle to be imaged as being reduced and projected on a workpiece (not shown) arranged on a workpiece table WT. The workpiece is a photosensitive substrate such as a semiconductor wafer on which a resist film is applied.

1.2 Operation of Exposure Apparatus **200**

[0036] The exposure apparatus **200** synchronously translates the reticle stage RT and the workpiece table WT in directions opposite to each other. Thus, the workpiece is exposed to the laser light LB reflecting the reticle pattern. Through the exposure process as described above, the reticle pattern is transferred onto the semiconductor wafer. Thereafter, an electronic device can be manufactured through a plurality of processes.

[0037] Instead of the exposure apparatus **200**, laser processing may be performed by causing the laser light LB output from the laser device **100** to enter a laser processing device (not shown).

1.3 Configuration of Laser Device **100**

[0038] FIG. **2** shows the configuration of the laser device **100** according to the comparative example. The laser device **100** includes an oscillator **10**, a beam splitter **11**, a measurement instrument **12**, a controller **20**, an internal power source **31**, and a display **32**.

[0039] The oscillator **10** is a device that includes a laser medium (not shown) and a laser resonator (not shown) and outputs the laser light LB. The beam splitter **11** is arranged on an optical path of the laser light LB, and transmits a part of the laser light LB at high transmittance and reflects the other part. The measurement instrument **12** is arranged on the optical path of the light reflected by the beam splitter **11**, and outputs measurement values such as the energy, the center wavelength, and the spectral line width of the light.

[0040] The controller **20** includes a processor **21**, a volatile memory **22**, a nonvolatile memory **23**, and a capacitor **24**, and controls the entire laser device **100** including the oscillator **10**. The controller **20** is specifically configured or programmed to perform various processes included in the present disclosure. The processor **21** is a processing device that executes various programs stored in the nonvolatile memory **23**. The volatile memory **22** is a storage device such as a random access memory (RAM) that serves as a temporary storage when the processor **21** operates. The nonvolatile memory **23** is a storage device such as a flash memory that stores various programs and data to be taken over before and after a power source is interrupted. The capacitor **24** is a device that stores electric energy when the internal power source **31** is on, and supplies power to the processor **21**, the volatile memory **22**, and the nonvolatile memory **23** when the internal power source **31** becomes off.

[0041] The internal power source **31** is a device that, when the laser device **100** is in operation, receives power supply from an external power source **300**, adjusts a voltage and converts the voltage into a direct-current voltage, and supplies power to each unit of the laser device **100**. When the laser device **100** is not in operation, power supply from the internal power source **31** to each unit of the laser device **100** is stopped. The internal power source **31** corresponds to the outside of the controller **20** in the present disclosure. The display **32** is a device that displays various images and characters in accordance with an image generation signal output from the controller **20**.

1.4 Operation of Laser Device **100**

[0042] The controller **20** controls the oscillator **10** based on a target value such as the target energy, the target center wavelength, and the target spectral line width received from the exposure apparatus **200** and a measurement value acquired from the measurement instrument **12**. While performing such control, the controller **20** stores, in the volatile memory **22**, a parameter including either or both of a setting value for operating hardware and software of the laser device **100** and a measurement value acquired from the measurement instrument **12**.

[0043] When a user or an operator of the laser device **100** stops the operation of the laser device **100**, or when power supply from the external power source **300** is interrupted due to power failure or the like, the internal power source **31** also becomes off, and power supply to the controller **20** is

stopped. When the power supply is stopped, data in the volatile memory **22** disappears. However, since the parameter stored in the volatile memory **22** contains information useful for subsequent operation of the laser device **100**, it is necessary to take a backup thereof. Therefore, when the internal power source **31** becomes off, backup to the nonvolatile memory **23** is performed using the power supplied from the capacitor **24** to each unit of the controller **20**. In the next operation of the laser device **100**, the previous parameter can be taken over by reading out data from the nonvolatile memory **23**.

1.5 Problem of Comparative Example

[0044] The capacitor **24** may deteriorate over time, and power supply capability from the capacitor **24** may be reduced. When the power supply capability is reduced and, for example, the power supply is stopped during backup, not only the backup process is not completed, but also a part of the data in the nonvolatile memory **23** may be damaged. Therefore, it is necessary to determine deterioration of the capacitor **24** and replace the capacitor **24** before failure occurs.

[0045] To determine deterioration of the capacitor **24**, it is conceivable to measure the output voltage of the capacitor **24**. However, providing a mechanism for measuring the output voltage of the capacitor **24** separately may lead to an increase in the size and cost of the controller **20**.

[0046] The embodiments described below relate to determining deterioration of the capacitor **24** in a simple device configuration.

2. Laser Device **100** Determining Deterioration Based on Duration Time of Power Supply from Capacitor **24**

2.1 Operation when Power Source Becomes Off

[0047] FIG. **3** is a flowchart showing operation of the processor **21** when the internal power source **31** becomes off in a first embodiment. The configurations of the laser device **100** and the controller **20** in the first embodiment are similar to those in the comparative example.

[0048] In **S1**, when the internal power source **31** becomes off, the processor **21**, the volatile memory **22**, and the nonvolatile memory **23** start operation with the power supplied from the capacitor **24**.

[0049] In **S2**, the processor **21** stores, in the nonvolatile memory **23**, the parameter of the laser device **100** stored in the volatile memory **22**. Up to this point, the first embodiment is similar to the comparative example.

[0050] In **S3**, the processor **21** stores data indicating the duration time of the power supply from the capacitor **24** in the nonvolatile memory **23**, and updates the data at every fixed time. The process of **S3** may be performed in parallel with storing the parameter in the nonvolatile memory **23** in **S2**, but is preferably performed after the process of **S2** is completed. The duration time stored in the nonvolatile memory **23** may be a time from when the internal power source **31** becomes off or a time from when storing the parameter in the nonvolatile memory **23** in **S2** is completed.

[0051] The data of the duration time is updated to a large numerical value as time elapses.

Updating the data is repeated until the power supply from the capacitor **24** necessary for the operation of the processor **21** is stopped.

[0052] Updating data means overwriting data, and the data of the duration time stored before the updating may be erased. When the power supply from the capacitor **24** is stopped, the data cannot be updated, and only the latest data remains in the nonvolatile memory **23**. Therefore, the data of the duration time stored in the nonvolatile memory **23** at the time when the power supply from the capacitor **24** is stopped indicates the power supply capability of the capacitor **24**.

[0053] The measurement accuracy of the duration time depends on the fixed time in **S3**. The fixed time is preferably 3 seconds or less, and may be, for example, 1 second. For example, the data of the duration time may be updated by adding 1 for each second with the value at the end of **S2** being 0, and in this case, corresponds to the data obtained by truncating a decimal in the number of seconds from the end of **S2** until the power supply is stopped.

[0054] When the power supply from the capacitor **24** is stopped, processing of the present

flowchart is ended.

[0055] FIG. 4 shows data of a part of the nonvolatile memory **23** in the first embodiment. The nonvolatile memory **23** includes a plurality of sectors SE. One sector SE includes a plurality of pages PA. The page PA is the smallest unit of data for writing and reading in the nonvolatile memory **23**. The sector SE is the smallest unit of data erasure in the nonvolatile memory **23**. That is, although writing and reading of data can be performed for each page PA, data erasure is performed for all pages PA included in one sector SE and cannot be performed for each page PA. Since data overwriting also involves data erasure, the sector SE is the smallest unit thereof.

[0056] In S3 of FIG. 3, among the plurality of sectors SE included in the nonvolatile memory **23**, a free sector in which various programs or the parameter stored in S2 is not stored is selected. The data of the duration time of the power supply may be stored in one page PA of the selected sector SE. Alternatively, the data of the duration time may be stored in a nonvolatile memory **23** different from the nonvolatile memory **23** in which the various programs or the parameter is stored.

2.2 Operation when Power Source Becomes on

[0057] FIG. 5 is a flowchart showing the operation of the processor **21** when the internal power source **31** becomes on in the first embodiment.

[0058] In S5, when the internal power source **31** becomes on, the processor **21**, the volatile memory **22**, and the nonvolatile memory **23** start operation with the power supplied from the internal power source **31**.

[0059] In S6, the processor **21** reads out the parameter of the laser device **100** stored at the previous off-time of the internal power source **31** from the nonvolatile memory **23**, and causes the laser device **100** to start operation.

[0060] In S7, the processor **21** starts storing the parameter of the laser device **100** in the volatile memory **22**. The parameter is stored in the volatile memory **22** every time the parameter is newly acquired during the operation of the laser device **100**.

[0061] In S8, the processor **21** reads out, from the nonvolatile memory **23**, the data indicating the duration time of the power supply from the capacitor **24** stored at the previous off-time of the internal power source **31**.

[0062] In S9, the processor **21** outputs the data indicating the duration time of the power supply from the capacitor **24**. The data indicating the duration time may be displayed on the display **32**, or may be stored as log data in the nonvolatile memory **23** or another storage device. A user or an operator of the laser device **100** may determine deterioration of the capacitor **24** based on the data indicating the duration time.

[0063] The processor **21** may determine deterioration of the capacitor **24** based on the data indicating the duration time. For example, the degree of deterioration or necessity of replacement of the capacitor **24** may be determined based on whether or not the duration time has become shorter than a threshold, whether or not the duration time has become equal to or less than a certain ratio with respect to the value of the capacitor **24** when it was new, or whether or not the duration time has become equal to or less than a certain ratio with respect to a design value. Further, by referring to the log data related to the duration time, the time when the replacement is necessary may be predicted from the transition of the duration time. The determination result of such deterioration may be displayed on the display **32**, or may be stored as log data in the nonvolatile memory **23** or another storage device.

[0064] After S9, the processor **21** ends processing of the present flowchart.

2.3 Effect

[0065] (1) According to the first embodiment, the controller **20** includes the processor **21**, the volatile memory **22**, the nonvolatile memory **23**, and the capacitor **24**, and controls the laser device **100**. The processor **21** stores the parameter of the laser device **100** in the volatile memory **22** while receiving power supply from the internal power source **31**. The processor **21** receives power supply from the capacitor **24** when the power supply from the internal power source **31** is stopped, and

stores, in the nonvolatile memory **23**, the parameter stored in the volatile memory **22** and the information indicating the duration time of the power supply from the capacitor **24**.

[0066] Accordingly, by storing the information indicating the duration time of the power supply from the capacitor **24** in the nonvolatile memory **23**, it is possible to determine the deterioration of the capacitor **24** with a simple device configuration without using a voltmeter.

[0067] (2) According to the first embodiment, the processor **21** stores the information indicating the duration time in the nonvolatile memory **23** after storing the parameter in the nonvolatile memory **23** is completed.

[0068] Accordingly, by giving priority to the backup of the parameter, the backup can be reliably completed before the power supply from the capacitor **24** is stopped.

[0069] (3) According to the first embodiment, the processor **21** repeats the operation of storing the information indicating the duration time in the nonvolatile memory **23** until the power supply from the capacitor **24** necessary for the operation of the processor **21** is stopped.

[0070] Accordingly, the power supply capability of the capacitor **24** can be known by repeating the storing until the power supply is stopped.

[0071] (4) According to the first embodiment, the processor **21** repeats the operation of sequentially updating the information indicating the duration time stored in the nonvolatile memory **23**.

[0072] Accordingly, it is possible to know an accurate duration time by repeating the updating to the latest data.

[0073] (5) According to the first embodiment, the processor **21** updates the information indicating the duration time at every fixed time.

[0074] Accordingly, it is possible to keep the measurement accuracy of the duration time constant.

[0075] (6) According to the first embodiment, the parameter includes any one of a setting value for operating the laser device **100** and a measurement value acquired from the measurement instrument **12** of the laser device **100**.

[0076] Accordingly, by reliably performing back up of the parameter including the set value or the measurement value, the laser device **100** can be operated by taking over the immediately preceding parameter at the next startup.

[0077] (7) According to the first embodiment, the processor **21** reads out the information indicating the duration time stored in the nonvolatile memory **23** when the power supply from the internal power source **31** is resumed.

[0078] Accordingly, it is possible to determine the deterioration of the capacitor **24** by reading out the duration time.

[0079] (8) According to the first embodiment, the processor **21** causes the display **32** to display the information indicating the duration time.

[0080] Accordingly, it is possible for a user or an operator to determine the deterioration of the capacitor **24** by showing the duration time to the user or the operator.

[0081] (9) According to the first embodiment, the processor **21** stores the information indicating the duration time as log data.

[0082] Accordingly, a user or an operator can confirm the duration time at a later date. Further, by recording the transition of the duration time, it is possible to accurately predict timing when replacement of the capacitor **24** is necessary.

[0083] (10) According to the first embodiment, the processor **21** determines the deterioration of the capacitor **24** based on the information indicating the duration time stored in the nonvolatile memory **23** when the power supply from the internal power source **31** is resumed.

[0084] Accordingly, it is possible to appropriately determine whether or not it is necessary to replace the capacitor **24** by determining the deterioration.

[0085] (11) According to the first embodiment, the processor **21** causes the display **32** to display the determination result of the deterioration of the capacitor **24**.

[0086] Accordingly, the determination result can be indicated to a user or an operator.

[0087] (12) According to the first embodiment, the processor **21** stores the determination result of the deterioration of the capacitor **24** as log data.

[0088] Accordingly, a user or an operator can confirm the determination result at a later date. Further, by recording the transition of the determination result of the deterioration, it is possible to accurately predict timing when replacement of the capacitor **24** is necessary.

[0089] In other respects, the first embodiment is similar to the comparative example.

3. Laser Device **100** Sequentially Storing Information Indicating Duration Time in Different Regions of Nonvolatile Memory **23**

3.1 Operation when Power Source Becomes Off

[0090] FIG. **6** is a flowchart showing the operation of the processor **21** when the internal power source **31** becomes off in a second embodiment. The configurations of the laser device **100** and the controller **20** in the second embodiment are similar to those in the comparative example. The processes of S1 and S2 are similar to those in the first embodiment.

[0091] After S2, in S4, the processor **21** stores data other than an initial value in different regions of the nonvolatile memory **23** at every fixed time. The different regions are, for example, different pages PA in the same sector SE. The process of S4 may be performed in parallel with storing the parameter in the nonvolatile memory **23** in S2, but is preferably performed after the process of S2 is completed. The process of S4 is repeated until the power supply from the capacitor **24** necessary for the operation of the processor **21** is stopped. When the power supply from the capacitor **24** is stopped, the process of the present flowchart is ended.

[0092] FIG. **7** shows data of a part of the nonvolatile memory **23** in the second embodiment. First, data of the selected sector SE, which is a free sector, is reset. For example, an initial value “1” is stored in all pages PA included in the selected sector SE. Next, in S4, data is stored in a different page PA at every fixed time. When the page PA containing data other than the initial value “1” is overwritten, the entire sector SE must be reset. However, when data is stored in the page PA containing the initial value “1”, other pages PA are not affected. The data to be stored is preferably a fixed value and, for example, a value “0” is stored. In FIG. **7**, the initial value “1” is changed to the value “0” in order from the page PA at the upper left end.

3.2 Operation when Power Source Becomes on

[0093] The operation of the processor **21** when the power source becomes on is substantially the same as that in FIG. **5**. However, in S8, the data of the duration time of the power supply is given in terms of the number of the pages PA in which the value “0” is stored. The longer the duration time, the greater the number of the pages PA in which the value “0” is stored. The processor **21** counts the number of the pages PA in which the value “0” is stored in the selected sector SE. In S9, the processor **21** calculates and outputs the duration time by multiplying the number of the pages PA in which the value “0” is stored by the length of the fixed time in S4 of FIG. **6**. If there is any page PA in which neither the initial value “1” nor the value “0” is stored, it can be determined that the page PA is damaged.

3.3 Effect

[0094] (13) According to the second embodiment, the processor **21** repeats the operation of sequentially storing data related to the information indicating the duration time in a different page PA of the nonvolatile memory **23**.

[0095] Accordingly, even when data in the n-th page PA is damaged as the power supply from the capacitor **24** is terminated while the data is being stored in the n-th page PA, the data in the n-1-th page PA remains, so that the information indicating the duration time can be reliably maintained.

[0096] (14) According to the second embodiment, the processor **21** stores data at every fixed time.

[0097] Accordingly, it is possible to keep the measurement accuracy of the duration time constant.

[0098] (15) According to the second embodiment, the processor **21** resets all the pages PA of at least one free sector included in the nonvolatile memory **23** being different from the sector in which the parameter is stored, and then stores the data in the free sector.

[0099] Accordingly, it is possible to store the information indicating the duration time without affecting the sector SE in which the parameter is stored.

[0100] (16) According to the second embodiment, the processor **21** acquires the duration time based on the number of the pages PA in which data other than the initial value is stored, when the power supply from the internal power source **31** is resumed.

[0101] Accordingly, since the number of the pages PA in which the data is stored increases as the duration time increases, the duration time can be accurately measured.

[0102] (17) According to the second embodiment, the processor **21** stores the data at every fixed time, and acquires the duration time based on the number of the pages PA in which the data is stored and the length of the fixed time when the power supply from the internal power source **31** is resumed.

[0103] Accordingly, since the data is stored at every fixed time, the accurate duration time can be known by multiplying the number of the pages PA in which the data is stored by the length of the fixed time.

[0104] (18) According to the second embodiment, the processor **21** repeats the operation of sequentially storing the same data in a different page PA of the nonvolatile memory **23**.

[0105] Accordingly, the processing is simplified because the data is the same. Further, it is easy to determine whether or not the data is damaged.

[0106] In other respects, the second embodiment is similar to the first embodiment.

4. Others

[0107] The description above is intended to be illustrative and the present disclosure is not limited thereto. Therefore, it would be obvious to those skilled in the art that various modifications to the embodiments of the present disclosure would be possible without departing from the spirit and the scope of the appended claims. Further, it would be also obvious to those skilled in the art that the embodiments of the present disclosure would be appropriately combined.

[0108] The terms used throughout the present specification and the appended claims should be interpreted as non-limiting terms unless clearly described. For example, terms such as “comprise”, “include”, “have”, and “contain” should not be interpreted to be exclusive of other structural elements. Further, indefinite articles “a/an” described in the present specification and the appended claims should be interpreted to mean “at least one” or “one or more.” Further, “at least one of A, B, and C” should be interpreted to mean any of A, B, C, A+B, A+C, B+C, and A+B+C as well as to include combinations of any thereof and any other than A, B, and C.

Claims

1. A controller comprising a processor, a volatile memory, a nonvolatile memory, and a capacitor, and configured to control a laser device, the processor being configured to store a parameter of the laser device in the volatile memory when receiving power supply from outside the controller, and when the power supply from outside is stopped, store, in the nonvolatile memory, the parameter stored in the volatile memory and information indicating a duration time of power supply from the capacitor while receiving the power supply from the capacitor.
2. The controller according to claim 1, wherein the processor stores the information indicating the duration time in the nonvolatile memory after storing the parameter in the nonvolatile memory is completed.
3. The controller according to claim 1, wherein the processor repeats operation of storing the information indicating the duration time in the nonvolatile memory until the power supply from the capacitor necessary for operation of the processor is stopped.
4. The controller according to claim 1, wherein the processor repeats operation of sequentially updating the information indicating the duration time stored in the nonvolatile memory.
5. The controller according to claim 4, wherein the processor updates the information indicating the

duration time at every fixed time.

6. The controller according to claim 1, wherein the parameter includes any one of a set value for operating the laser device and a measurement value acquired from a measurement instrument of the laser device.

7. The controller according to claim 1, wherein the processor reads out the information indicating the duration time stored in the nonvolatile memory when the power supply from outside is resumed.

8. The controller according to claim 7, wherein the processor causes a display to display the information indicating the duration time.

9. The controller according to claim 7, wherein the processor stores the information indicating the duration time as log data.

10. The controller according to claim 1, wherein the processor determines deterioration of the capacitor based on the information indicating the duration time stored in the nonvolatile memory when the power supply from outside is resumed.

11. The controller according to claim 10, wherein the processor causes a display to display a determination result of the deterioration of the capacitor.

12. The controller according to claim 10, wherein the processor stores a determination result of the deterioration of the capacitor as log data.

13. The controller according to claim 1, wherein the processor repeats operation of sequentially storing data related to the information indicating the duration time in a different region of the nonvolatile memory.

14. The controller according to claim 13, wherein the processor stores the data at every fixed time.

15. The controller according to claim 13, wherein the processor resets all regions of at least one free sector included in the nonvolatile memory being different from a sector in which the parameter is stored, and then stores the data in the free sector.

16. The controller according to claim 13, wherein the processor acquires the duration time based on a number of regions in which the data is stored when the power supply from outside is resumed.

17. The controller according to claim 13, wherein the processor stores the data at every fixed time, and when the power supply from outside is resumed, acquires the duration time based on a number of regions in which the data is stored and a length of the fixed time.

18. The controller according to claim 1, wherein the processor repeats operation of sequentially storing same data in a different region of the nonvolatile memory.

19. A laser device comprising an oscillator configured to output laser light, and a controller configured to control the oscillator, the controller including a processor, a volatile memory, a nonvolatile memory, and a capacitor, and the processor being configured to store a parameter of the laser device in the volatile memory when receiving power supply from outside the controller, and when the power supply from outside is stopped, store, in the nonvolatile memory, the parameter stored in the volatile memory and information indicating a duration time of power supply from the capacitor while receiving the power supply from the capacitor.

20. An electronic device manufacturing method, comprising: generating laser light using a laser device; outputting the laser light to an exposure apparatus; and exposing a photosensitive substrate to the laser light in the exposure apparatus to manufacture an electronic device, the laser device including an oscillator configured to output laser light, and a controller configured to control the oscillator, the controller including a processor, a volatile memory, a nonvolatile memory, and a capacitor, and the processor being configured to store a parameter of the laser device in the volatile memory when receiving power supply from outside the controller, and when the power supply from outside is stopped, store, in the nonvolatile memory, the parameter stored in the volatile memory and information indicating a duration time of power supply from the capacitor while receiving the power supply from the capacitor.
